Statistical Shape Modeling of the Femur

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Introduction to Statistical Shape Modeling

A modeling algorithm's role is to generate a representation of an anatomical structure. In this project, we will perform the statistical shape modeling of the femur bone.

Here, the objective is to create a Statistical Shape Model of the femur bone based on 22 point clouds representing the patients' femurs.

Creating the Project Environment

The language used was Python version 3.10.9, and the libraries used were: re, time, numpy, scipy.spatial (KDTree), pyvista, os, pycpd (DeformableRegistration), time.

The point clouds are imported using a custom script with regex recognition. The data are stored in the format of Open Inventor. Here is a description of the initial dataset by number of points:

Table 1: Number of Vertices for Each Femur Mesh

Femur ID	**Point Count**	**Femur ID**	**Point Count**
Femur 1	66,100	Femur 11	68,261
Femur 2	48,515	Femur 12	71,273
Femur 3	73,521	Femur 13	62,754
Femur 4	66,651	Femur 14	76,055
Femur 5	75,746	Femur 15	60,362
Femur 6	72,103	Femur 16	65,991
Femur 7	67,317	Femur 17	69,168
Femur 8	57,168	Femur 18	60,309
Femur 9	56,928	Femur 19	55,772
Femur 10	82,841	Femur 20	78,863

I chose femur 10 (Python index 9) as the template, as it contains the largest number of points in its cloud.

Down-sampling

The resolution of the cloud points (median = 66984.0; min = 48515 ; max = 82841) is excessively high for modeling purposes. Due to the high computational complexity of CPD, we must only select a fraction of these points. Thus, before performing the successive steps of modeling, the femur point clouds are down-sampled.

At first, I chose the points at random in the point clouds. This caused a massive loss of resolution at the extremities, thus degrading the information available about the shape of the femoral head.

Afterward, I decided to use the Farthest Point Algorithm. The principle of the algorithm is to maximize the sum of Euclidean distances between the points in the selected subset : $f(S) = \min_{S \subset E} \sum_{(x,y) \in S^2} ||x - y||^2$

Here is a description of the Farthest Point Algorithm for an initial cloud point E comprising N points and a subset selection S of m points.

- One point is chosen at random to initialize the selection. At each moment in the execution, a point is in one of two states: either selected in the subset or remaining.
- At each step, we select one of the remaining points until we reach a subset of m selected points. The point is chosen as follows:
  - For every remaining point, we find its nearest neighbor among the selected points and calculate its distance to it (topologically, this is the distance of a point to a subset). Our objective is to maximize the sum of distances, so we select the point whose distance to the nearest neighbor is maximal.

I also reduced the number of points sampled from 4000 to 1000 to run the Deformable Registration faster with CPD.

Rigid Registration

The first step is to perform the Scaled Rigid Registration of the 20 femurs. The objective is to align each femur into the same standard coordinate system. The result is a transformation, providing a rotation, translation and scaling for each femur to fit the reference. In this registration, the shape is maintained, only the position and scale change.

In the transformation, the fixed points are the template reference femur, and the moving points are those of each femur. The Procrustes principle detailed in the three points below was applied in the icp_scaled function. The "while" loop ended after a maximum of 50 iterations OR when the mean error was below the 1e-5 threshold.

- Initial Step: Choose one femur as a "Static" (Reference) surface and align all 20 femurs to it.
- Refinement Step: Calculate the Mean Shape of all aligned femurs.
- Iteration: Make this new Mean Shape the "Static" surface and re-align all 20 femurs to this mean. Repeat until the mean stabilizes.

  In this process, to obtain each alignment during the iterations, we compute a Similarity Transformation : Rotation R, Translation t, and Scale s.

Step 1 is implemented using KDTree in the icp_scaled function.

Steps 2 through 8 are implemented in the compute_similarity_transform.

1. Correspondences: For every point in the Moving cloud, find the closest point in the Static cloud (using KDTree).
2. Centroids: Calculate the center of mass of both the Moving points and their corresponding Static points.
3. Center: Subtract centroids to center both clouds at the origin $(0, 0, 0)$.
4. Covariance: Compute the cross-covariance matrix $H = \text{MovingCentered}^T \cdot \text{StaticCentered}$.
5. SVD: Perform Singular Value Decomposition on H: $[U, S, V^T] = \text{np.linalg.svd}(H)$.
6. Rotation (R): $R = V \cdot U^T$. (Check if determinant is -1 to handle reflections).
7. Scale (s): Calculated using the ratio of variances of the centered points or the trace of the SVD scaling factors.
8. Translation (t): $t = \text{StaticCentroid} - s \cdot (R \cdot \text{MovingCentroid})$.

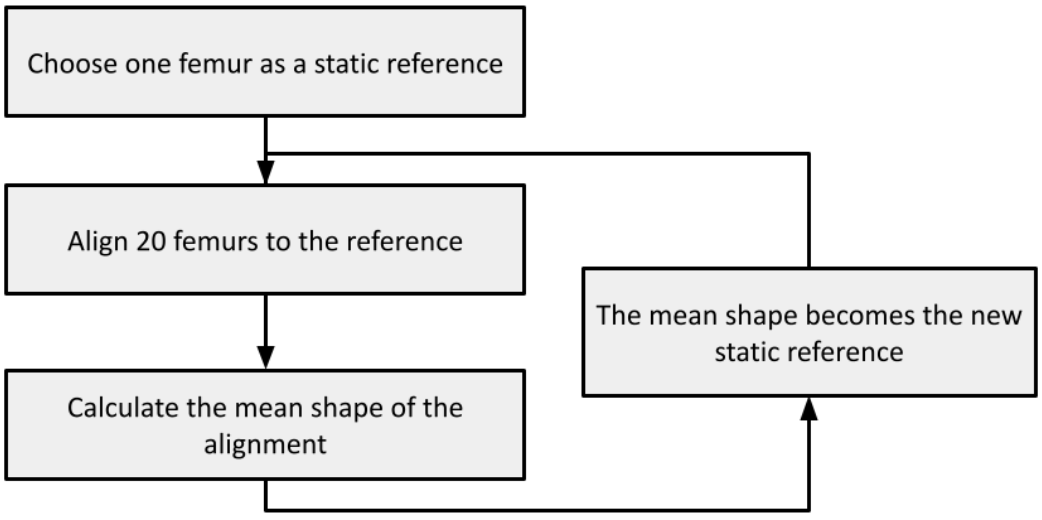


Figure 1: General Procrustes Algorithm

Deformable Registration

This step consists of applying deformable point registration through the coherent point drift (CPD) algorithm. This time, it consists of finding the transformation that allows the template (moving points) to fit each femur (fixed points).

The result of this step is the acquisition of transformation matrices. This step proved to be extremely costly computationally. The parameters that influence the duration of execution are the number of femurs, size of point clouds (set at 1000), alpha (set at 3.0), and beta (set at 2.0). We arrived at a total execution time of 8 minutes and 17 seconds.

With more patience and computing power (HPC cluster), we could rerun our pipeline using a higher number of points, and assess the influence of downsampling on the model's performance.

Principal Component Analysis

Principal Component Analysis is used to assess the variability of the femurs obtained after both registration steps.

For the PCA, the femurs are each represented by a vector of 3000 coordinates: their 1000 $\times$ 3 coordinate matrix is concatenated into a 3000 coordinate vector in section 3.1. Then they are stacked in a 20 row matrix (data_matrix).

Our data has been prepared for PCA, by the two previous transformations:

- The Rigid Registration allows us to discard the pose (position and rotation): the patient's position during acquisition does not matter.
- The Deformable Registration creates a one-to-one mapping which is necessary to ensure the comparability of the data: Column 1 is always the x-coordinate of the tip of the femoral head, and Column 3,000 is always the z-coordinate of the lateral condyle. Without the deformable registration, the one-to-one correspondence is lost, and the PCA no longer has signification.

As a dimension reduction tool, PCA allows us to better understand the variations between different femur types. These variations are captured in "modes" of variation: they correspond to the eigenvectors of highest eigenvalue obtained through the SVD. We can visualize, in the Results and visualizations sections, how the variability occurs in the dataset according to the two first modes.

Results and visualizations

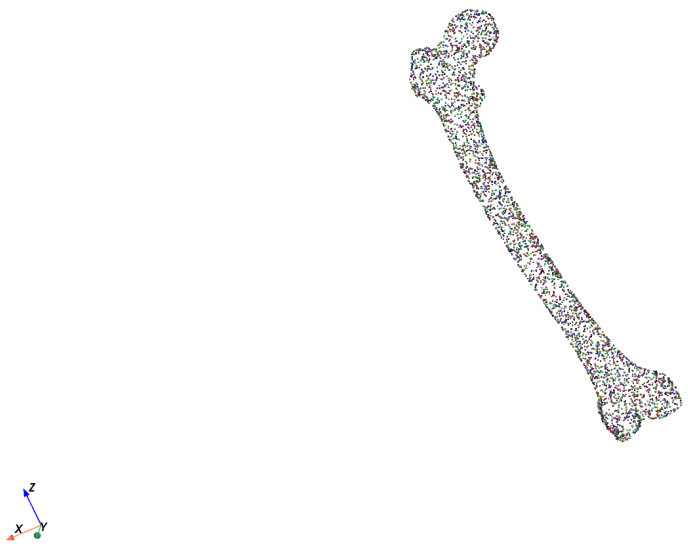


Figure 2: Chosen template for registration (femur 10), full resolution

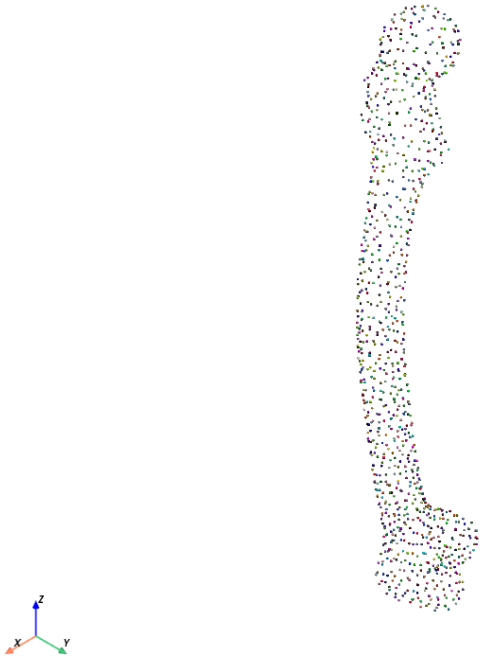


Figure 3: Chosen template for registration (femur 10), downsampled to 1000 points

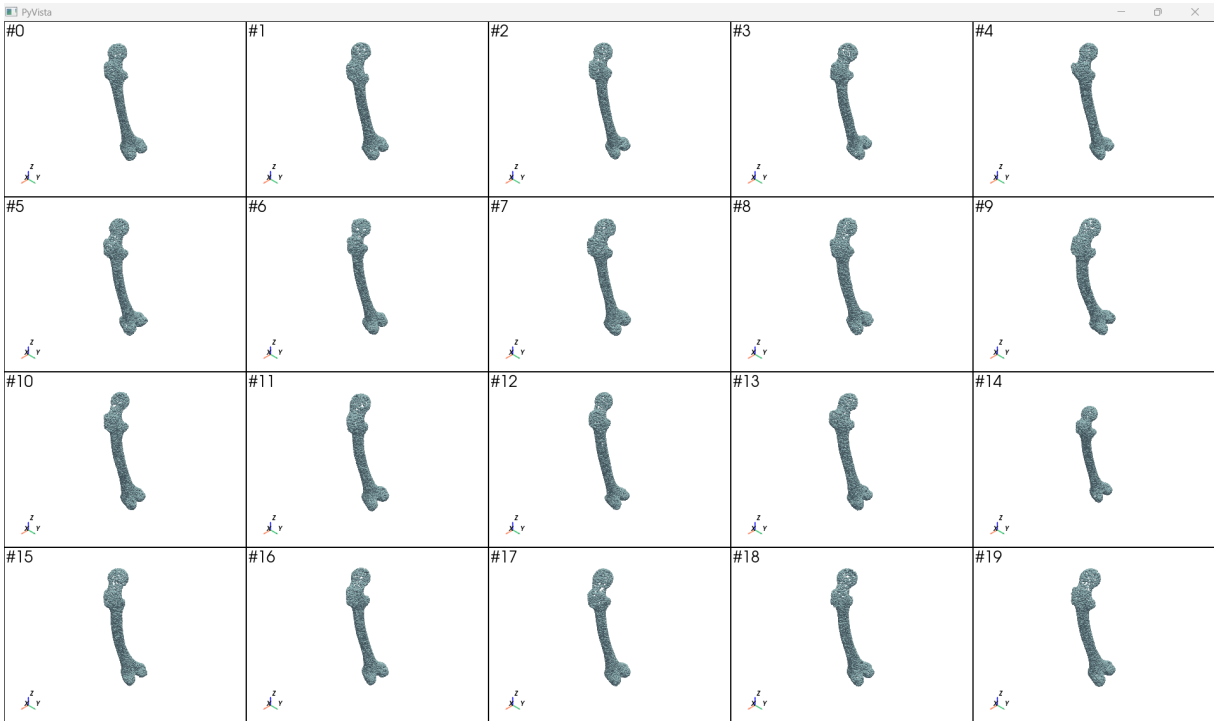


Figure 4: ICP Rigid Registration with original point clouds

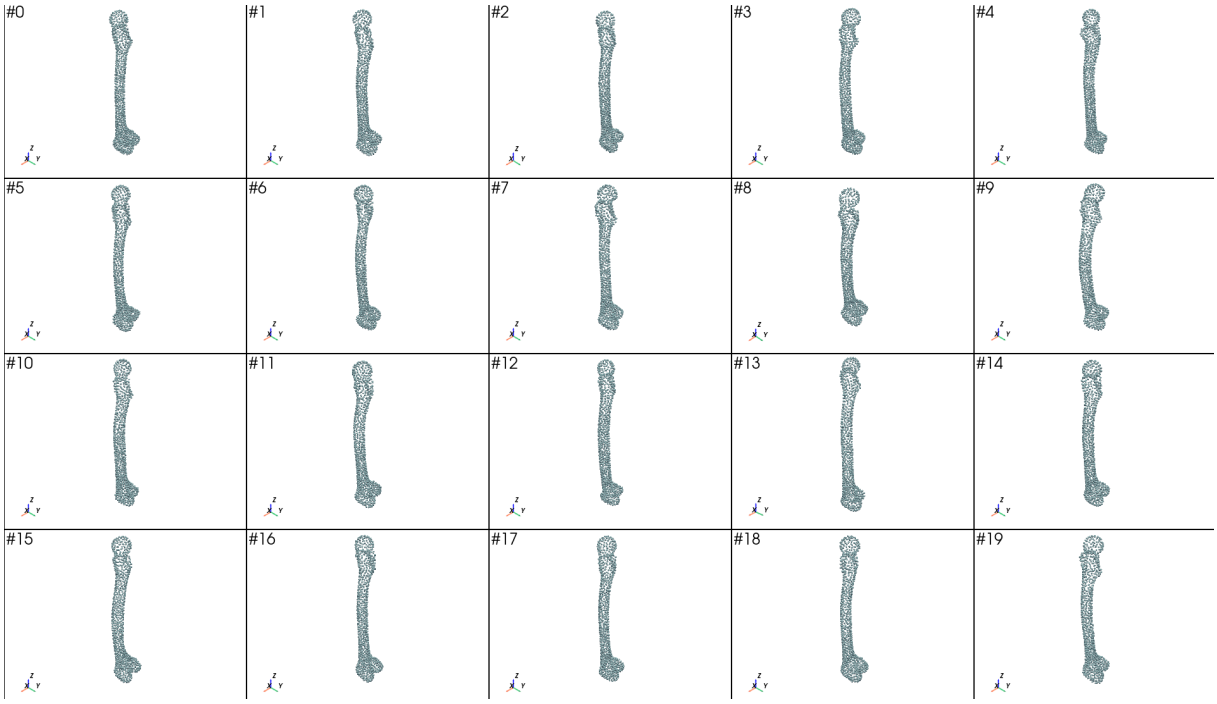


Figure 5: ICP Rigid Registration downsampled to 1000 points

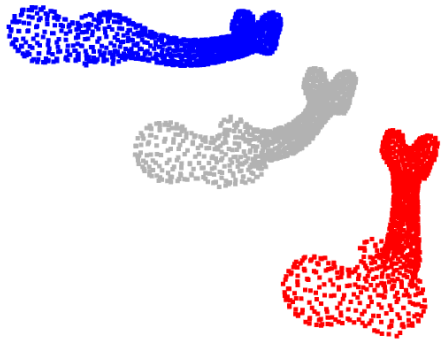


Figure 6: Mode 1: Left (Blue): -3 SD — Middle (Gray): Mean — Right (Red): +3 SD

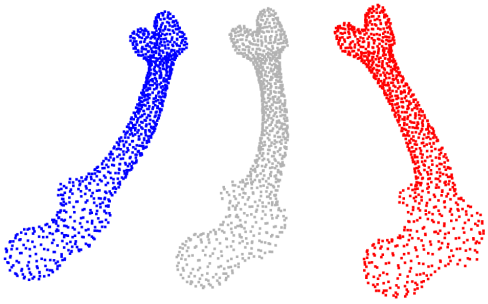


Figure 7: Mode 2 : Left (Blue): -3 SD — Middle (Gray): Mean — Right (Red): +3 SD

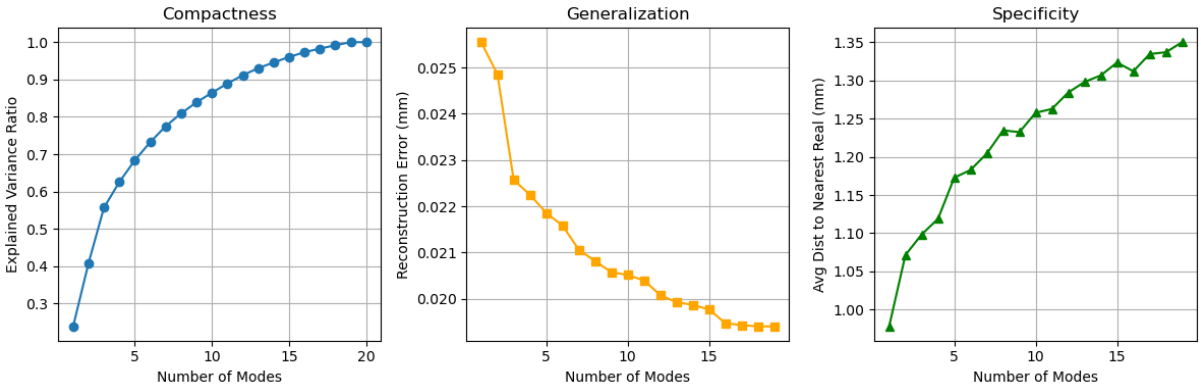


Figure 8: Evaluation Metrics for the SSM

Principal Component Analysis revealed distinct modes of anatomical variation. Mode 1 accounted for the largest variance: it corresponds to the length and thickness of the femoral head relative to the shaft (as seen in Fig. 6). Mode 2 captured shape differences in the proximal femur, specifically variations in neck thickness and inclination (Fig. 7).

The quantitative metrics validate the model's performance. The generalization error converged to a low residual value (0.02 model units), showing high adaptability to new shapes, while Compactness analysis indicated that 15 modes were sufficient to represent 95% of the dataset's variability.

Others



Figure 9: Triangle mesh for visualization

In Fig. 9, I tried to visualize the femur "modes" using computed triangle meshes instead of the cloud points. The rendering was failed, which led me to use the cloud points.

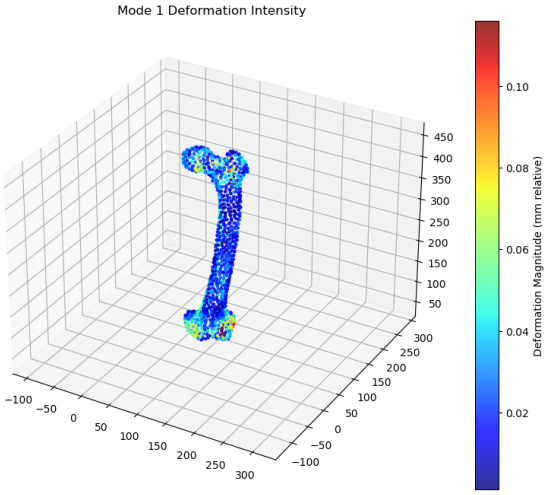


Figure 10: Mode 1 Deformation Intensity

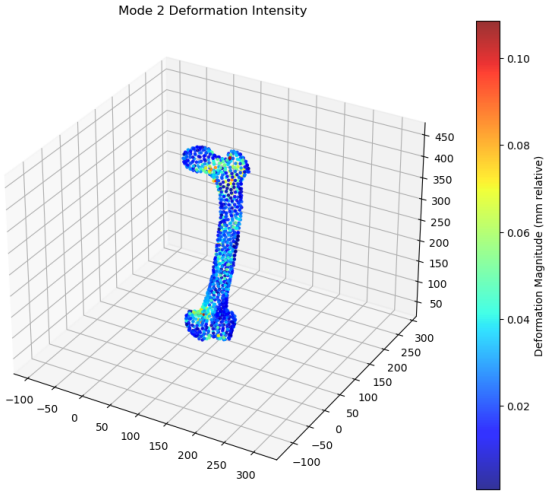


Figure 11: Mode 2 Deformation Intensity

For a given eigenvector (or mode), the graphs above plot the mean shape colored by the magnitude of the eigenvector at a given coordinate. Red represents a high deformation area for the given mode, whereas blue represents a low deformation area. We find that most of the variability occurs at the femoral extremities, especially at the head, the neck and the condyle.